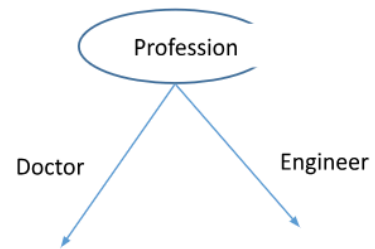


CSE 427 Fall 25
Class Test 1
Time 30 Minutes

Section 3
Student ID:

Find the root of the left branch of the given regression tree

Dosage	Profession	Drug effectiveness4
6	Doctor	0
8	Engineer	2
10	Doctor	3
12	Engineer	4
14	Doctor	5
16	Engineer	100
20	Doctor	100
25	Engineer	100
30	Doctor	100
35	Engineer	40
40	Doctor	30
45	Engineer	20
50	Doctor	5



Brac University
CSE427: Machine Learning
Fall 2025
Quiz 01

Duration: 30 minutes

Name:	ID:	Section: 06
-------	-----	-------------

Question 1:

Find the appropriate node at the leftmost branch of the regression tree along with predicted values of the leaves using the data given in the following table.

Store Type	Promotion	Season	Sales(\$)
Urban	Discount	Summer	72
Urban	Buy1Get1	Summer	70
Urban	Discount	Winter	68
Urban	Discount	Summer	74
Urban	Buy1Get1	Winter	65
Suburban	Discount	Summer	82
Suburban	Discount	Winter	78
Suburban	Buy1Get1	Summer	80
Suburban	Buy1Get1	Winter	76
Suburban	Discount	Summer	84
Rural	Discount	Summer	52
Rural	Discount	Winter	48
Rural	Buy1Get1	Winter	46
Rural	Discount	Summer	54
Rural	Buy1Get1	Summer	50

